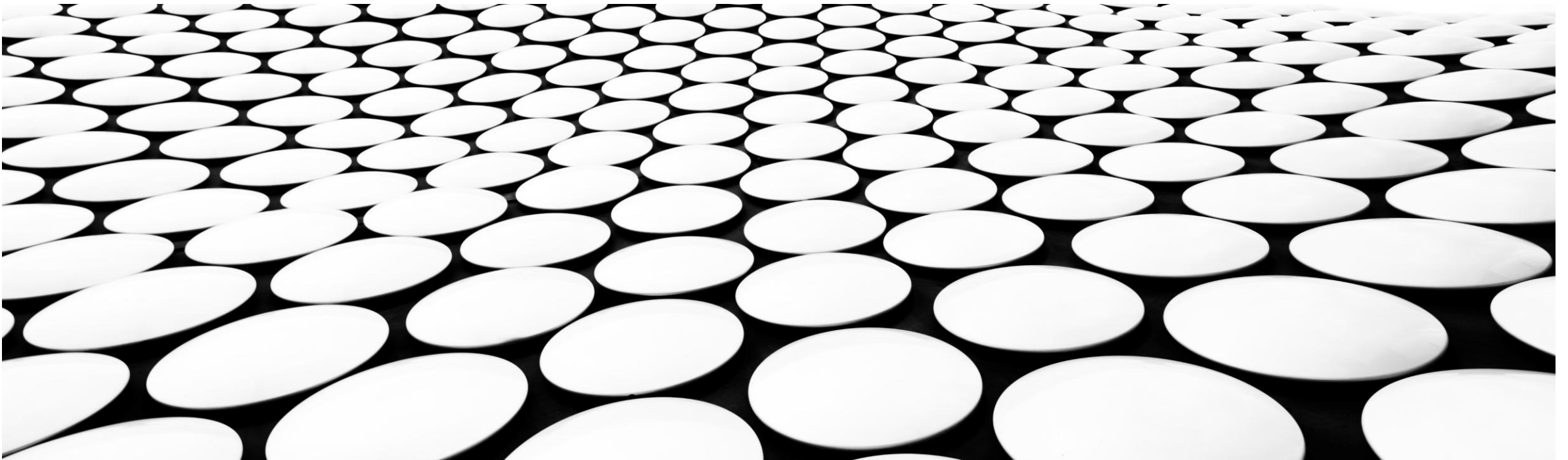

UNCOVERING THE DRIVERS BEHIND HIGH-VALUE ORDERS

EXPLORATORY DATA ANALYSIS (EDA) - PROJECT 2

BY- SHASHWAT PANDA



PROBLEM STATEMENT

Our objective is to diagnose the underlying patterns in our sales data to understand what drives maximum revenue, who our VIP customers are, and whether our current promotional discount codes are actually incentivizing larger purchases.





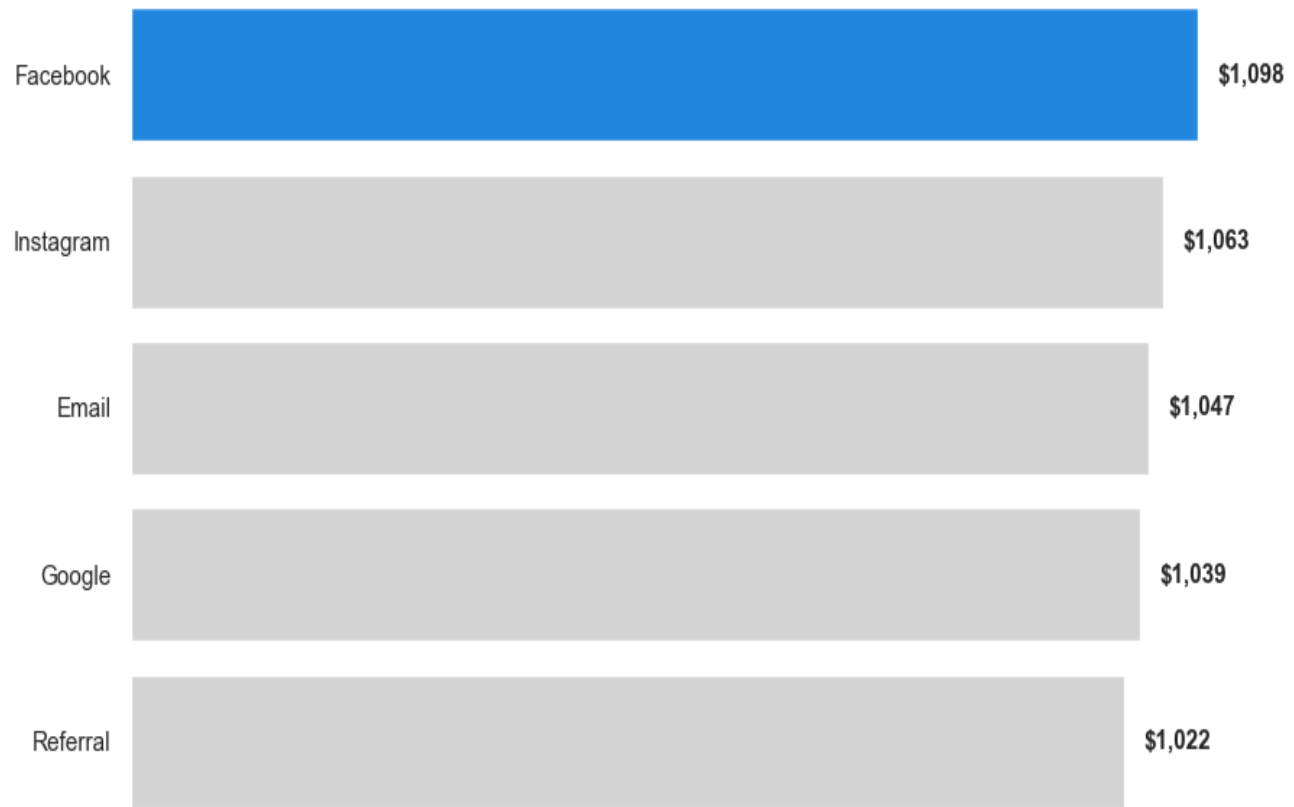
A FORENSIC APPROACH TO DATA INTEGRITY

- **Data Cleansing:** Sanitized the raw digital evidence by isolating invalid date formats and standardizing promotional code text to eliminate "Noise".
- **Locating the Centre:** Calculated the "centre of gravity" for business revenue using the Five-Number Summary, identifying a right-skewed distribution.
- **Signal Detection:** Deployed the Interquartile Range (IQR) method—which is robust against extreme values—to mathematically isolate VIP outliers from standard organic traffic.

FACEBOOK REFERRALS AND BULK ELECTRONICS DRIVE MAXIMUM REVENUE

- **The VIP Profile:** The top 1% of our highest-spending outlier orders are driven almost entirely by maximum-quantity purchases (5 items) of high-ticket electronics (Monitors, Laptops, Tablets).
- **Social Media Dominance:** Facebook referrals generate the highest average total spend (\$1,098), outperforming Instagram (\$1,062) and Google (\$1,039).
- **Coupon Inefficiency:** Current promotional codes (WINTER15, SAVE10) do not increase cart sizes. Customers using no coupon ("None") actually averaged the highest cart size (5.56 items).

Facebook Referrals Drive the Highest Average Spend





STRATEGIC PIVOTS FOR MARKETING AND PROMOTIONS

- **Target B2B Bulk Buyers:** Since our highest revenue comes exclusively from maximum-quantity electronics orders, we should pivot ad spend toward B2B clients or office managers upgrading their tech.
- **Double-Down on Facebook:** Reallocate ad spend away from lower-performing channels and concentrate marketing efforts on Facebook, our most profitable referral source.
- **Restructure Discounts:** Phase out flat-rate coupons like SAVE10. Implement threshold-based incentives (e.g., "15% off orders of 6 or more items") to actively force an increase in average cart sizes.